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Long-term exposure to ambient air pollution and road traffic noise and asthma incidence in adults: The Danish Nurse cohort

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ABSTRACT

Background: Ambient air pollution is likely a risk factor for asthma, and recent evidence suggests the possible relevance of road traffic noise.

Objectives: We examined the associations of long-term exposure to air pollution and road traffic noise with adult-asthma incidence.

Methods: We followed 28,731 female nurses (age > 44 years) from the Danish Nurse Cohort, recruited in 1993 and 1999, for first hospital contact for asthma from 1977 until 2015. We estimated residential annual mean concentrations of particulate matter with diameter < 2.5 µm (PM_{2.5}) since 1990 and nitrogen dioxide (NO₂) since 1970 with the Danish DEHM/UBM/AirGIS modeling system, and road traffic noise (L_{den}) since 1970 with the Nord2000 model. Time-varying Cox regression models were used to associate air pollution and road traffic noise exposure with asthma incidence.

Results: During 18.6 years' mean follow-up, 528 out of 23,093 participants had hospital contact for asthma. The hazard ratios (HR) and 95% confidence intervals for asthma incidence associated with 3-year moving average exposures were 1.29 (1.03, 1.61) per 6.3 µg/m³ for PM_{2.5}, 1.16 (1.07, 1.27) per 8.2 µg/m³ for NO₂, and 1.12 (1.00, 1.25) per 10 dB for L_{den}. The HR for NO₂ remained unchanged after adjustment for either PM_{2.5} or L_{den}, while the HRs for PM_{2.5} and L_{den} attenuated to unity after adjustment for NO₂.

Conclusions: Long-term exposure to air pollution was associated with adult-asthma incidence independently of road traffic noise, with NO₂ most relevant. Road traffic noise was not independently associated with adult-asthma incidence.

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1. Introduction

Asthma is a common chronic disease characterized by respiratory symptoms and reversible airflow limitation (Papi et al. 2018). Genetic and environmental factors contribute to the complex etiology of asthma development, with ambient air pollution increasingly recognized as a possible risk factor (Gilmour et al. 2006; Guarneri and Balmes 2014). The evidence supportive of associations between early-life air pollution exposure and asthma and asthma-like symptoms in children continues to grow (>40 studies) (Gehring et al. 2020; Khreis et al. 2017), while literature on adult-asthma incidence is more modest (twelve studies), mainly due to the lack of cohorts with information on lifelong asthma incidence (Jacquemin et al. 2012; Thurston et al. 2020). Given that adult asthma seems to be mechanistically different from childhood asthma and is associated with other risk factors, a separate focus on this specific phenotype is important (Trivedi and Denton 2019). Although literature is suggestive of an association between long-term exposure (≥ 1 year to lifetime) to air pollution and adult-asthma incidence, uncertainties still remain about the shape of the concentration–response function, especially at low-level exposures, as well as which air pollutants are most relevant, both important for disease prevention and air pollution regulation.

Another possibly important environmental stressor relevant for asthma is noise. In Europe, noise from traffic (road, railway, and air) was ranked as the third major environmental risk factor for burden of disease, after particulate matter with diameter < 2.5 μm ($\text{PM}_{2.5}$) and secondhand smoke (Hänninen et al. 2014). While evidence on associations between road traffic noise and cardiovascular diseases has been established (Kempen et al. 2018; Ndrepepa and Twardella 2011), literature on respiratory outcomes is just emerging, with two studies considering asthma in adults (Cai et al. 2017; Eze et al. 2018); one suggested associations with asthma symptom exacerbation (Eze et al. 2018), but not with asthma prevalence (Cai et al. 2017) or incidence (Eze et al. 2018). A study in children found no associations with asthma incidence or prevalence (Wallas et al. 2020). A proposed biological pathway for the association of noise with asthma may be through noise-induced psychological stress and/or noise induced sleep disturbances that have been associated with asthma morbidity, possibly by immune modulation (Recio et al. 2016b; Yonas et al. 2012). Thus, the association between road traffic noise and adult asthma warrants further investigation (Linares and Díaz 2019; Recio et al. 2016a). Road traffic noise and ambient air pollutants, such as nitrogen dioxide (NO_2), share major source - traffic, making it imperative to consider potential mutual confounding or interactive effects of each other.

We aimed to investigate the single and joint effects of long-term exposure to ambient air pollution and road traffic noise on adult-asthma incidence in a nationwide cohort in Denmark.

2. Methods

2.1. Study population

The Danish Nurse Cohort, inspired by the American Nurses' Health Study, was initiated in 1993, by including female nurses who were > 44 years old and members of the Danish Nurse Organization (comprising 95% of Danish nurses) (Hundrup et al. 2012). In brief, 19,898 out of 23,170 invited female nurses answered the initial questionnaire in 1993. In 1999, 489 non-responders from 1993 and 8,344 new nurses who turned 44 years between 1993 and 1999 were additionally included. Here we used the baseline information from questionnaires in 1993 (19,898 nurses) or 1999 (8,833 nurses), which included lifestyles (e.g. smoking, physical activity), education, marital status, and occupational characteristics. The establishment of the cohort was approved by the Scientific and Ethical Committee of Copenhagen and Frederiksberg Municipalities, and the Danish Data Protection Agency.

2.2. Outcome definition

Using a unique personal identification number given to all persons living in Denmark, we linked the cohort to the Danish Civil Registration System to obtain the information of residential address history and date of death or emigration since 1970 until December 31, 2014 or August 10, 2015, respectively. We linked the cohort to the Danish National Patient Register to extract information on the hospital contacts (inpatient, outpatient, and emergency room visit) for asthma between 1977 and August 10, 2015, using primary discharge diagnoses of International Classification of Diseases (ICD), 8th version: 493 (up to 1993) and 10th version: J45–46 (from 1994). We defined incidence of asthma as first hospital contact for asthma between the cohort baseline in 1993 or 1999 and August 10, 2015, after excluding participants with asthma hospital contact between 1977 and baseline.

2.3. Exposure assessment

2.3.1. Outdoor air pollution

Annual mean concentrations of $\text{PM}_{2.5}$ and particulate matter with diameter < 10 μm (PM_{10}) from 1990, and NO_2 and nitrogen oxide (NO_x) from 1970 until 2014 at residential addresses of the participants were modeled with the Danish DEHM/UBM/AirGIS modeling system (<https://au.dk/airgis/>) (Jørgen Brandt et al. 2001a; Brandt et al., 2001c; Brandt et al. 2003; Brandt et al. 2012; Jensen et al. 2001; Jensen et al. 2017). The modelling system is covering a wide range of spatial scales at three different levels (regional background, urban background, street contribution), and therefore consists of the three following submodules. The Danish Eulerian Hemispheric Model (DEHM) is applied for the long-range transported components with a nested grid set-up covering the northern hemisphere, calculates the regional background concentrations in a 5.6 km $\times$ 5.6 km grid resolution. (Brandt et al. 2012; Frohn et al. 2002). The Danish Urban Background Model (UBM) is applied for the urban background in a 1 km $\times$ 1 km grid resolution covering Denmark (Jørgen Brandt et al. 2001a; Brandt et al., 2001c; Jørgen Brandt et al. 2001b; Brandt et al. 2003). The Operational Street Pollution Model (OSPM) models the concentration level for streets contributions with >500 vehicles per day; and AirGIS generates input files automatically for OSPM to construct a dataset with modelled concentrations at residential address level (Berkowicz 2000; Ketzel et al. 2012). The necessary data for carrying out the exposure modelling also included emissions database for Denmark and meteorological datasets (e.g. wind speed, direction, air temperature) (Khan et al. 2019). The DEHM/UBM/AirGIS modeling system has been recently evaluated for its performance for the four air pollutants ($\text{PM}_{2.5}$, PM_{10} , NO_2 , and NO_x) by comparing with monitor measurements, showing a range of correlations of 0.45–0.96 in temporal variation and 0.32–0.92 in spatial variation (Khan et al. 2019). As for the wide range of spatial correlation, the lowest two values of were observed for $\text{PM}_{2.5}$ (0.32) and PM_{10} (0.62), which were based on only 20 data points and therefore contained relative uncertainty being sensitive to single outliers; and the highest value was observed for NO_2 (0.92) (Khan et al. 2019). In this study, we had available annual mean concentrations of $\text{PM}_{2.5}$ and PM_{10} from 1990, and NO_2 and NO_x from 1970 until 2014 at residential addresses of the participants. We calculated the time-weighted annual mean level at the year of one address change as the mean level at both two addresses, using the exact date of moving.

2.3.2. Road traffic noise

The road traffic noise levels at residence were calculated using the Nord2000 propagation model (DELTA 2001), with following input variables: address geocodes, the heights of residential buildings, road lines with information on annual average daily traffic, road traffic composition and speed, road types and properties (e.g., motorway, rural highway, roads wider than 6 m, and other roads), building polygons for all surrounding buildings (including building heights), and meteorology

(including wind speed/direction, air temperature, and cloud coverage). Based on the geometrical ray theory, the propagation model computes the one-third octave band sound attenuation following the path from the sound source to its receiver. It accounts for the properties of terrain, such as shape, ground type, impedance, and roughness, as well as variations in weather patterns, which are appropriate when calculating annual average noise levels and needed to be predefined. The Nord2000 method has been validated extensively (DELTA 2006, 2009). Specifically, nine cases with predicted annual L_{den} levels by the Nord2000 propagation model from a road, covering propagation distances up to 300 m, showed an average difference < 0.5 dB and a standard uncertainty < 1 dB comparing with the measurements (DELTA 2006).

We estimated the annual mean levels of residential road traffic noise for all participants from 1970 until 2014, as the equivalent A-weighted sound pressure level (LA_{eq}) at the most exposed façade, estimating noise contributions from roads within a 3-km radius from the addresses. Annual average road traffic noise levels were calculated separately for day (L_d ; 07:00–19:00 h), evening (L_e ; 19:00–22:00 h), night (L_n ; 22:00–07:00 h), and the overall weighted 24-h noise level throughout the day, evening, and night adding a penalty of 5 dB to the evening hours and 10 dB to the night hours (L_{den}). Addresses located in remote areas secluded from traffic noise contributions might be returned by the Nord2000 model as 0 dB. However, noise levels of 0 dB never occur in reality because of noise contributions from other sources (e.g., vegetation). We set the noise levels of 0 dB (with no road traffic noise contributions) to 35 dB (accounting for $< 2\%$ of total addresses), assuming that this level would correspond to the average background noise contribution.

2.4. Statistical analysis

We used time-varying Cox regression models with age as the underlying timescale (Thiebaut and Benichou 2004) to examine the associations of long-term exposures to air pollution and road traffic noise with asthma incidence, with censoring at the time of death, emigration, or the end of follow-up in 2015, whichever came first. Air pollution and road traffic noise exposures were included as linear variables. The associations were first estimated in two single-pollutant models: 1) the crude model only adjusted for age (time axis), year of baseline (1993, 1999), and calendar year of follow-up (strata one year); 2) the fully-adjusted model further adjusted for *a priori* defined covariates, including smoking status (never, previous, current), smoking duration (years), smoking intensity (grams/day), body mass index (BMI, categorical variable according to WHO: < 18.5 , 18.5 – 24.9 , 25.0 – 29.9 , ≥ 30 kg/m²), marital status (married, separated, divorced, single, widowed), occupational status (working, homeworking, retired, unemployed, other), leisure-time physical activity status (high, medium, low), vegetable consumption (daily, weekly, rarely), avoidance of fatty meat (yes, no), and alcohol consumption [never, moderate (1–14 drinks/week), heavy (> 15 drinks/week)].

The missing exposure values ($\sim 4.2\%$) for certain years for a participant were interpolated using the averages of available exposure values of other years. For example, we substituted the missing $PM_{2.5}$ annual value in 1990 by the average of other non-missing annual values from 1991 to 2014 for a participant. Given that the cut-off years for exposure assessment and cohort follow-up were 2014 and 2015, respectively, we matched each year of follow-up with the annual mean exposure value for a year prior to that follow-up year. We applied four different exposure time windows: 1-, 3-, 10-, and 23-year moving averages (arithmetic means for air pollutants and geometric means for road traffic noise). Exposure windows of 3- and 23-year were the longest available exposure windows for $PM_{2.5}$ and PM_{10} (since 1990) and NO_2 , NO_x , and L_{den} (since 1970), respectively. Analyses with 3-year moving average were considered as the main analyses, as this is the longest exposure window for all exposures available. The results are presented as hazard ratio (HR) and 95% confidence interval (CI) for an interquartile range (IQR)

increase for each air pollutant and per 10 dB increase for L_{den} (close to IQR = 9.4).

We also fitted mutually adjusted models (two- or three-pollutant) based on the fully-adjusted model, in order to estimate the mutually independent associations and disentangle which exposures are most relevant to asthma incidence. Due to high correlations, we did not run two-pollutant models with $PM_{2.5}$ and PM_{10} ($r = 0.90$) or NO_2 and NO_x ($r = 0.92$). We also calculated the joint effects and sought to understand the effects of multipollutant exposures on asthma incidence based on mutually adjusted models (Crouse et al. 2015; Jerrett et al. 2013; Klompaker et al. 2019; Wang et al. 2019). The HRs for the joint effects were calculated for an IQR increase for each pollutant and a 10 dB for L_{den} , as the exponential sum of the product of each exposure's coefficient in the mutually adjusted model and that exposure's IQR. Standard errors for joint effects were calculated using the variance-covariance matrices for the mutually adjusted exposures' coefficients (Crouse et al. 2015; Jerrett et al. 2013).

To examine if associations persisted at lower air pollution concentrations, we re-ran the single and mutually adjusted models by excluding observations with above pre-defined cut-off values: 25, 20, 15 $\mu\text{g}/\text{m}^3$ for $PM_{2.5}$; and 40, 30, 20, 15, 10 $\mu\text{g}/\text{m}^3$ for NO_2 . We additionally estimated association with L_{den} keeping only observations with $L_{den} > 48$ or 53 dB and with categorized L_{den} (low/ < 48 dB; medium/48–53 dB; and high/ > 53 dB). To evaluate the shapes of concentration-response curves, we applied natural cubic splines with two degrees of freedom in both single and mutually adjusted analyses. We compared the model fits of linear and spline models using the likelihood-ratio test to assess the deviations from linearity.

We performed effect modification of the associations for $PM_{2.5}$, NO_2 , and L_{den} exposures by baseline year (1993 or 1999), age ($<$ or ≥ 65 years), obesity (BMI ($<$ or ≥ 30 kg/m²), smoking status, physical activity, marital status, occupational status, urbanization (rural, < 180 persons/km²; provincial, 180–5,220 persons/km²; urban, $> 5,220$ persons/km²), and categorized $PM_{2.5}$, NO_2 , and L_{den} exposures, by introducing an interaction term into the main model and comparing interaction and main in model the performing the Wald test.

We conducted several sensitivity analyses to examine the robustness of associations: 1) estimated associations with air pollutant and noise levels at the year before baseline and baseline year exposures by fixed Cox regression model; 2) compared alternative approaches for model adjustment by calendar year of follow-up: without adjustment, a linear term, natural cubic splines, strata for 1-year, and strata for 5-years; 3) estimated the effects from adjusting for different groups of covariates, and 4) excluded observations with missing values for exposures for certain years during follow-up.

All statistical analyses were performed using R (version 3.6.1). Exposure maps were created using features of the Spatial Analyst extension to Esri's ArcMap (version 10.7.1; Esri).

3. Results

Of the 28,731 participants, we excluded 4 with inactive status before baseline, 223 with asthma before baseline, 860 with missing exposure information, and 4,551 with missing information on covariates, leaving 23,093 participants for analyses. During a mean follow-up of 18.6 years, 528 participants had first hospital contact for asthma (Table 1). The mean age at baseline was 52.8 (range 43.9–91.0) years. Participants who developed asthma during the follow-up were more likely to be overweight or obese, smokers, live in urban areas and have higher air pollution levels at the baseline addresses than participants without asthma (Table 1 and Table 2).

Pearson correlations between road traffic noise and air pollutants varied between 0.19 and 0.61 (Table S1). We observed high correlations between different time-windows for the same exposure (0.79–0.99). The correlations between different exposures did not change over time (Table S2). We observed higher concentrations of $PM_{2.5}$ in southeastern

Table 1

Descriptive statistics at the baseline year (1993/1999) for the Danish Nurse Cohort study participants.

Characteristics	Total N = 23,093	No asthma N = 22,565	Asthma N = 528	P- values*
Year of enrollment, N (%)				
1993	14,964 (65)	14,593 (65)	371 (70)	0.01
1999	8,129 (35)	7,972 (35)	157 (30)	
Follow-up time, years (Mean ± SD)	18.6 ± 4.7	18.8 ± 4.4	10.0 ± 6.2	<0.001
Age, years (Mean ± SD)	52.8 ± 7.8	52.8 ± 7.8	52.2 ± 7.0	0.08
Age < 65 years old, N (%)	21,021 (91)	20,532 (91)	489 (93)	0.23
BMI, kg/m ² , (Mean ± SD)	23.7 ± 3.5	23.7 ± 3.5	24.4 ± 4.1	<0.001
BMI, categories, N (%)				
Underweight (<18.5 kg/m ²)	564 (2)	551 (2)	13 (2)	<0.001
Normal weight (18.5–24.9 kg/ m ²)	15,961 (69)	15,629 (69)	332 (63)	
Overweight (25.0–29.9 kg/ m ²)	5,250 (23)	5,120 (23)	130 (25)	
Obese (≥30 kg/m ²)	1,318 (6)	1,265 (6)	53 (10)	
Smoking status, N (%)				
Never	8,279 (36)	8,099 (36)	180 (34)	0.54
Previous	6,960 (30)	6,790 (30)	170 (32)	
Current	7,854 (34)	7,676 (34)	178 (34)	
Smoking duration, years (Mean ± SD)	10.2 ± 15.1	10.2 ± 15.1	10.7 ± 16.0	0.43
Smoking intensity, grams/day (Mean ± SD)	4.6 ± 7.9	4.6 ± 7.9	5.4 ± 9.0	0.03
Marital status, N (%)				
Married	16,408 (71)	16,048 (71)	360 (68)	0.08
Separated	395 (2)	382 (2)	13 (2)	
Divorced	2,586 (11)	2,511 (11)	75 (14)	
Single	2,238 (10)	2,195 (10)	43 (8)	
Widowed	1,466 (6)	1,429 (6)	37 (7)	
Occupational status, N (%)				
Working	18,436 (80)	18,006 (80)	430 (81)	0.50
Homeworking	411 (2)	399 (2)	12 (2)	
Retired	3,871 (17)	3,790 (17)	81 (15)	
Unemployed	153 (1)	152 (1)	1 (0.2)	
Other	222 (1)	218 (1)	4 (1)	
Leisure time physical activity, N (%)				
High	6,223 (27)	6,085 (27)	138 (26)	0.36
Medium	15,384 (67)	15,036 (67)	348 (66)	
Low	1,486 (6)	1,444 (6)	42 (8)	
Vegetable consumption, N (%)				
Daily	17,139 (74)	16,749 (74)	390 (74)	0.77
Weekly	5,611 (24)	5,485 (24)	126 (24)	
Rarely	343 (1)	331 (1)	12 (2)	
Avoids fatty meat, N (%)	20,948 (91)	20,460 (91)	488 (92)	0.20
Alcohol consumption, N (%)				
Never (0 drinks/week)	3,449 (15)	3,358 (15)	91 (17)	0.85
Moderate (1–15 drinks/week)	14,330 (62)	14,028 (62)	302 (57)	
Heavy (>15 drinks/week)	5,314 (23)	5,179 (23)	135 (26)	

Table 1 (continued)

Characteristics	Total N = 23,093	No asthma N = 22,565	Asthma N = 528	P- values*
Urbanization level, N (%)†				<0.001
Rural	9,548 (41)	9,359 (41)	189 (36)	
Provincial	10,029 (44)	9,805 (44)	224 (42)	
Urban	3,516 (15)	3,401 (15)	115 (22)	

SD: standard deviation; BMI: body mass index.

*: P-values for difference of no asthma and asthma groups were obtained from one-way analysis of variance (one-way ANOVA) and Pearson's Chi-square test.

†: Urbanization level was defined by population density at municipality of residence at the cohort baseline year (1993/1999): rural areas: <180 persons/km²; provincial areas: 180–5,220 persons/km²; and urban areas: 5,220 persons/km².

Denmark, while NO₂, NO_x, and L_{den} were higher in urban areas (Fig. 1 and Figure S1). PM₁₀ concentrations were higher along the west and east coast lines. There was a consistently decreasing trend for all air pollutants during the study period, while road traffic noise levels remained stable (Figure S2).

In single-pollutant models, we observed positive and significant associations between 3-year moving average exposures to PM_{2.5}, NO₂, NO_x, and L_{den}, not PM₁₀, and asthma incidence, with small attenuations of estimates in the fully adjusted models (Table 3). Consistent associations were detected for other exposure windows of 1-, 10-, and 23-year moving average exposure (Table S3 and Table S4).

The HRs for the associations between PM_{2.5}, NO₂, and L_{den} and asthma incidence were more pronounced at lower concentrations of air pollutants or at higher levels of noise (Table 3, Table S3, and Table S4), but there was no deviation from linearity ($p > 0.05$) and no evidence for a threshold (Fig. 2). In mutually adjusted models, the HRs for NO₂ remained unchanged after adjusting for PM_{2.5} or L_{den}, while the HRs for PM_{2.5} and L_{den} were attenuated to unity (Table 3, Table S3, Table S4, and Figure S3).

In three-pollutant models, the HR for NO₂ remained unchanged, whereas the HRs for PM_{2.5} and L_{den} were attenuated to close-to-unity (Fig. 3). The combined HRs assuming independent for PM_{2.5}, NO₂, and L_{den} were slightly higher than the corresponding joint HRs, with no interaction effects between these exposures (Table S5). In addition, estimates in the three joint effect models with NO₂ (HRs = 1.16, 1.17, and 1.17) were almost equal to the estimate in the single model for NO₂ (HR = 1.16) (Fig. 3).

Observed associations were robust when using different exposure periods (Table S6), different approaches of adjustment for calendar year (Figure S4 and Table S7), different covariate adjustment methods, or exclusion of imputed exposure data for analyses (Table S8). We found no effect modification of observed associations by any of factors explored (Table S9).

4. Discussion

We found positive associations between long-term exposure to ambient air pollution, especially NO₂ which is a proxy for traffic-related air pollution, and adult-asthma incidence, independently of road traffic noise. The associations persisted at low exposure levels, with no threshold observed. We also present a novel finding of an association between road traffic noise and adult-asthma incidence, however, the association was attenuated to null in mutually adjusted models with NO₂.

Our results expand the literature supporting associations between air pollution and adult-asthma incidence, in particular with respect to exposure-response relationships at low levels and traffic-derived air

Table 2

Characteristics of residential air pollution and road traffic noise levels at the year before baseline for 23,093 participants from the Danish Nurse Cohort.

Exposures	Total N = 23,093			No asthma N = 22,565			Asthma N = 528			P-values*
	Mean ± SD	IQR	Range	Mean ± SD	IQR	Range	Mean ± SD	IQR	Range	
Air pollution										
PM _{2.5} , µg/m ³	18.94 ± 4.12	6.33	5.73–46.39	18.92 ± 4.11	6.35	5.73–46.39	19.69 ± 4.18	6.02	9.46–41.27	<0.001
PM ₁₀ , µg/m ³	22.63 ± 4.55	7.02	9.13–56.29	22.61 ± 4.55	7.03	9.13–56.29	23.31 ± 4.50	6.32	11.36–49.12	<0.001
NO ₂ , µg/m ³	12.79 ± 7.89	8.19	2.55–78.76	12.76 ± 7.85	8.15	2.55–78.76	13.98 ± 9.22	9.08	3.82–71.38	<0.001
NO _x , µg/m ³	19.97 ± 23.53	12.17	2.70–434.98	19.89 ± 23.37	12.13	2.70–434.98	23.29 ± 29.39	12.97	4.80–325.47	0.001
Road traffic noise										
L _{den} , dB	52.66 ± 8.11	9.40	5.30–79.50	52.64 ± 8.11	9.33	5.40–79.50	53.52 ± 8.17	10.03	5.30–73.90	0.01
L _{den} , N (%)†										
Low	5,127 (22)	—	—	5,031 (22)	—	—	96 (18)	—	—	0.08
Medium	6,412 (28)	—	—	6,257 (28)	—	—	155 (29)	—	—	
High	11,554 (50)	—	—	11,277 (50)	—	—	277 (53)	—	—	
L _{24h} , dB	48.58 ± 8.08	9.30	5.10–75.20	48.56 ± 8.08	9.30	5.10–75.20	49.48 ± 7.96	9.69	10.00–69.90	0.01
L _d , dB	50.41 ± 8.15	9.30	5.80–77.10	50.39 ± 8.15	9.30	5.80–77.10	51.32 ± 8.00	9.59	11.90–72.00	0.01
L _e , dB	48.05 ± 8.05	9.30	5.20–74.60	48.03 ± 8.06	9.30	5.20–74.60	48.96 ± 7.93	9.64	9.30–69.10	0.01
L _n , dB	44.53 ± 7.87	9.30	5.40–71.50	44.51 ± 7.87	9.30	5.40–71.50	45.38 ± 7.82	10.02	5.70–66.20	0.01

IQR: interquartile; dB: decibels; PM_{2.5}: particulate matter with an aerodynamic diameter of <2.5 µm; PM₁₀: particulate matter with an aerodynamic diameter of <10 µm; NO₂: nitrogen dioxide; NO_x: nitrogen oxide; L_{24h}: L_{24hour}; L_d: L_{day}; L_e: L_{evening}; L_n: L_{night}. 53 dB is the current WHO environmental noise guideline limit value for the European region for exposure to outdoor road traffic noise.

*: P-values for difference of no asthma and asthma groups were obtained from one-way analysis of variance (one-way ANOVA) and Pearson's chi-square test. —, not applicable.

†: Three levels of categorized L_{den} are low (<48 dB), medium (48–53 dB), and high (>53 dB).

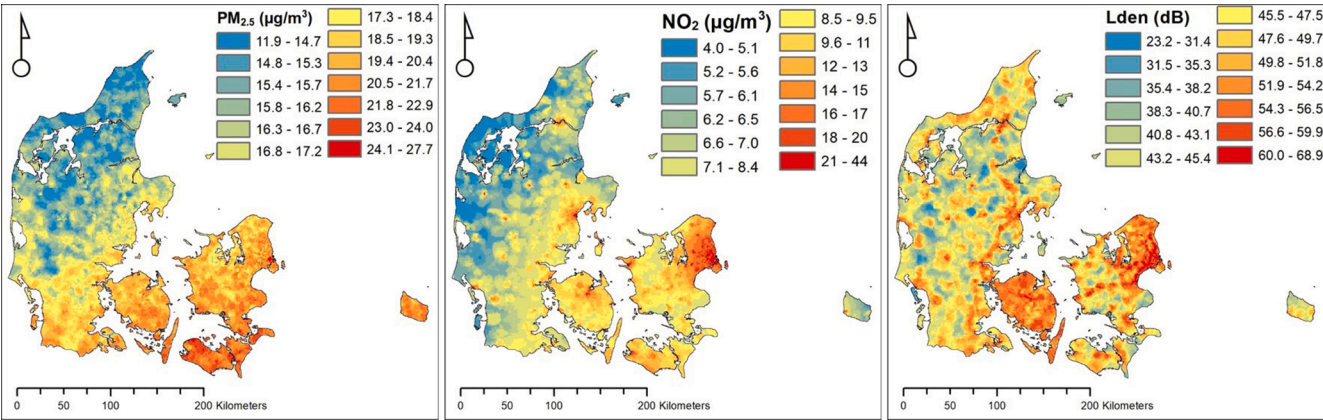


Fig. 1. Smoothed maps of estimated annual residential exposure levels for PM_{2.5}, NO₂, and L_{den} in the Danish Nurse Cohort at the year before baseline (1992 or 1998). Smoothed pollution maps at 1 × 1 km spatial resolution were based on 23,093 participants' residential exposure estimates at the year before baseline. We applied ordinary kriging method using spherical semivariogram model based on a variable search radius with data from 12 nearest participants.

pollution as relevant culprit. Among the twelve published studies, six out of seven studies with PM_{2.5} (Hendryx et al. 2019; Jacquemin et al. 2015; Mostafavi et al. 2019; Salimi et al. 2018; Weichenthal et al. 2017; Young et al. 2014) and all eight studies with NO₂ (Andersen et al. 2012; Jacquemin et al. 2009; Jacquemin et al. 2015; Modig et al. 2009; Mostafavi et al. 2019; Salimi et al. 2018; Weichenthal et al. 2017; Young et al. 2014) found positive associations. The Danish Diet, Cancer and Health study (Andersen et al. 2012) and the Canadian study in Toronto (Weichenthal et al. 2017) reported significant positive associations between NO₂ and adult-asthma incidence defined by hospital contacts as in our study, with the Canadian study also showing a significant association for PM_{2.5} (Weichenthal et al. 2017). A cohort study in Sydney, Australia, observed non-significant positive associations of PM_{2.5} and NO₂ with asthma hospitalization (Salimi et al. 2018). Our findings also generally agree with studies using self-reported asthma diagnoses. Jacquemin et al. in a study among 17 European cities (Jacquemin et al. 2009) and Modig et al. in a Swedish study (Modig et al. 2009) found significant positive associations with NO₂. In a Swiss cohort study of never-smokers, Künzli et al. reported significant positive associations of PM₁₀ with adult-asthma incidence (Künzli et al. 2009). In the European Study of Cohorts for Air Pollution Effects (ESCAPE) study, Jacquemin

et al. found positive but non-significant associations for PM_{2.5}, PM₁₀, NO₂, and NO_x (Jacquemin et al. 2015). In the American Sisters Cohort Study Young et al. found positive but non-significant associations with PM_{2.5} and NO₂ (Young et al. 2014). Another two studies from Switzerland and Australia also reported positive non-significant associations for PM_{2.5}, PM₁₀, and NO_x/NO₂ (Hendryx et al. 2019; Mostafavi et al. 2019). On the contrary, the American Nurses' Health Study of 121,701 female nurses reported no associations of PM_{2.5} and PM₁₀ exposures with asthma incidence (Fisher et al. 2016).

Our results agree with a single other study on road traffic noise and adult-asthma incidence (Eze et al. 2018). In a Swiss study, Eze et al. found that transportation noise and road traffic noise were not associated with either prevalence or incidence of asthma, but were associated with asthma symptoms/exacerbation among asthmatics, independently of NO₂ (Eze et al. 2018). A cross-sectional study found no association of road traffic noise with asthma prevalence (Cai et al. 2017). A Swedish study of children with longitudinal follow-up from before birth to 16 years old found no association between residential road traffic noise exposure and incident or prevalent asthma (Wallas et al. 2020). In our study, we observed a marginally significant association for L_{den}, while after adjustment for NO₂, the association was attenuated to null. Based

Table 3Associations of 3-year moving average (MA) air pollution and road traffic noise L_{den} with asthma incidence in main and subset analyses in the Danish Nurse Cohort.

	N cases	Crude*	Fully-adjusted†	Fully-adjusted† + PM _{2.5}	Fully-adjusted† + NO ₂	Fully-adjusted† + L _{den}
		HR (95% CI)	HR (95% CI)	HR (95% CI)	HR (95% CI)	HR (95% CI)
3-year MA						
PM _{2.5}	528	1.35 (1.09, 1.68)	1.29 (1.03, 1.61)	—	0.99 (0.74, 1.35)	1.21 (0.95, 1.54)
PM ₁₀	528	1.21 (0.97, 1.51)	1.16 (0.92, 1.45)	—	0.89 (0.68, 1.17)	1.09 (0.86, 1.38)
NO ₂	528	1.18 (1.08, 1.28)	1.16 (1.07, 1.27)	1.17 (1.03, 1.32)	—	1.16 (1.04, 1.29)
NO _x	528	1.07 (1.03, 1.11)	1.06 (1.02, 1.11)	1.05 (0.99, 1.11)	—	1.05 (1.00, 1.11)
PM_{2.5}						
< 25 µg/m ³	520	1.39 (1.09, 1.78)	1.32 (1.03, 1.70)	—	1.04 (0.76, 1.42)	1.25 (0.95, 1.63)
< 20 µg/m ³	447	1.63 (1.18, 2.24)	1.55 (1.12, 2.14)	—	1.20 (0.81, 1.78)	1.47 (1.04, 2.07)
< 15 µg/m ³	220	1.58 (0.81, 3.08)	1.47 (0.75, 2.88)	—	1.56 (0.71, 3.41)	1.50 (0.75, 3.03)
NO₂						
< 40 µg/m ³	521	1.26 (1.13, 1.40)	1.24 (1.11, 1.38)	1.24 (1.08, 1.43)	—	1.26 (1.10, 1.45)
< 30 µg/m ³	510	1.33 (1.17, 1.52)	1.32 (1.15, 1.51)	1.33 (1.13, 1.57)	—	1.36 (1.15, 1.62)
< 20 µg/m ³	468	1.37 (1.14, 1.65)	1.35 (1.12, 1.63)	1.38 (1.11, 1.72)	—	1.45 (1.15, 1.83)
< 15 µg/m ³	386	1.43 (1.08, 1.88)	1.43 (1.08, 1.89)	1.48 (1.09, 2.01)	—	1.49 (1.08, 2.07)
< 10 µg/m ³	239	0.91 (0.47, 1.77)	0.90 (0.46, 1.74)	0.94 (0.45, 1.97)	—	0.84 (0.40, 1.75)
L_{den} (continuous)						
All levels	528	1.14 (1.02, 1.27)	1.12 (1.00, 1.25)	1.08 (0.96, 1.22)	1.01 (0.88, 1.15)	—
L _{den} > 48 dB	432	1.17 (0.98, 1.40)	1.14 (0.95, 1.36)	1.07 (0.88, 1.30)	0.94 (0.75, 1.19)	—
L _{den} > 53 dB	302	1.23 (0.95, 1.58)	1.21 (0.93, 1.56)	1.08 (0.82, 1.43)	0.96 (0.69, 1.32)	—
L_{den} (categorical)						
Low (<48 dB)	96	Ref.	Ref.	Ref.	Ref.	—
Medium (48–53 dB)	130	1.08 (0.83, 1.41)	1.09 (0.83, 1.41)	1.07 (0.82, 1.39)	1.05 (0.80, 1.36)	—
High (>53 dB)	302	1.23 (0.98, 1.55)	1.20 (0.95, 1.52)	1.13 (0.89, 1.44)	1.04 (0.81, 1.34)	—

Estimates are presented as hazard ratios (HRs) and 95% confidence intervals (CIs) based on IQR increases for PM_{2.5} (6.33 µg/m³), PM₁₀ (7.02 µg/m³), NO₂ (8.19 µg/m³), and NO_x (12.17 µg/m³), and per 10 dB increase for L_{den}.

*: Crude model, adjusted for age (time axis), year of baseline (1993/1999), and calendar year of follow-up (strata one year);

†: Fully-adjusted model, further adjusted for smoking (status, duration, and intensity), BMI (category), marital status, occupational status, leisure time physical activity status, vegetable consumption (category), avoids fatty meat, and alcohol consumption (category).

25 µg/m³ for PM_{2.5} and 40 µg/m³ for NO₂ are the current European air quality limit values. 53 dB is the current WHO environmental noise guideline limit value for the European region for exposure to outdoor road traffic noise.

—, not applicable in two pollutant models due to the high correlation of two pollutants (above 0.7).

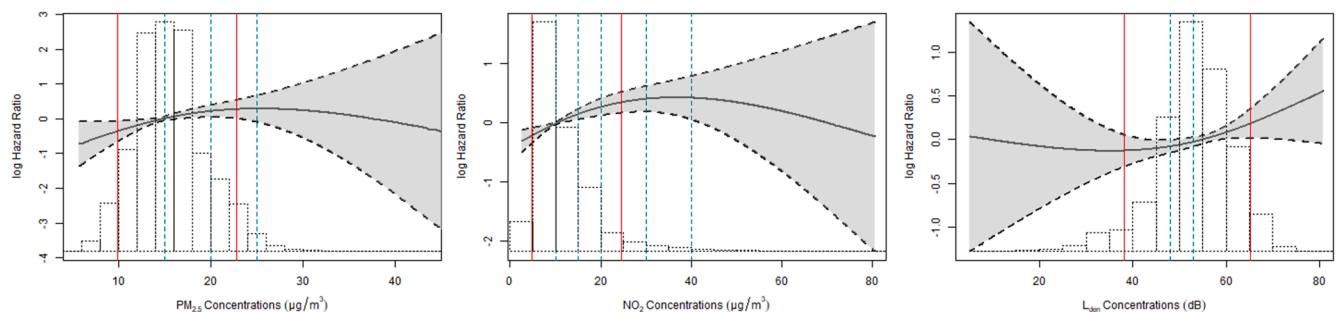


Fig. 2. Natural cubic splines with two degrees of freedom of the associations between 3-year moving averages PM_{2.5}, NO₂, and L_{den} exposures and asthma incidence in the Danish Nurse Cohort. The curve results were based on fully-adjusted models. Black solid lines indicate log hazard ratio values and black dashed lines indicate their 95% confidence intervals. Red solid lines indicate the 5th and 95th percentiles of levels for PM_{2.5}, NO₂, and L_{den}. Green dashed lines indicate the cut-off values for PM_{2.5} (25, 20, and 15 µg/m³), NO₂ (40, 30, 20, 15, and 10 µg/m³), and L_{den} (53 and 48 dB). The histograms show the distributions of 3-year moving averages of PM_{2.5}, NO₂, and L_{den}. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

on limited existing evidence, there are some suggestions of road traffic noise being a possible stressor in exacerbating symptoms in subjects with asthma, but no consistent support exists for road traffic noise as a risk factor for asthma development.

Ambient air pollution is a complex mixture of particulate and gaseous pollutants generated from multiple sources, and it is highly relevant to identify specific components relevant for health, for regulation and prevention purposes (Tétreault et al. 2013). Few studies on air pollution and adult-asthma incidence investigated the independent associations of multiple air pollutants in mutually adjusted models (Jacquemin et al. 2015; Weichenthal et al. 2017). The ESCAPE study found similar tendencies for NO₂ and PM₁₀ that were comparable with our results (Jacquemin et al. 2015), reporting a robust association for NO₂ with a slightly increased HR (1.10 vs. 1.17) after adjusting for PM₁₀, while the HR for PM₁₀ attenuated

(1.04 vs. 0.98) after adjusting for NO₂. In our study, the HR for NO₂ increased from 1.16 to 1.20 after PM₁₀ adjustment, while the HR for PM₁₀ decreased from 1.16 to 0.89 after NO₂ adjustment. Weichenthal et al. found that positive association of both PM_{2.5} and NO₂ with asthma in single-pollutant models, persisted in mutually adjusted models (Weichenthal et al. 2017). We observed that the HR for PM_{2.5} attenuated to unity in models with NO₂ (1.29 vs. 0.99), whereas the HR for NO₂ remained unchanged after PM_{2.5} adjustment (1.16 vs. 1.17), suggesting that NO₂, a proxy of traffic-related air pollution, is the most relevant pollutant for asthma etiology. We cannot, however, conclude whether the association with NO₂ reflects direct effect of NO₂ itself or, to some extent, other closely correlated traffic combustion pollutants, such as black carbon, and ultra-fine particles, calling for more detailed research to address this important question. In addition, the pollutant with the lowest measurement error may

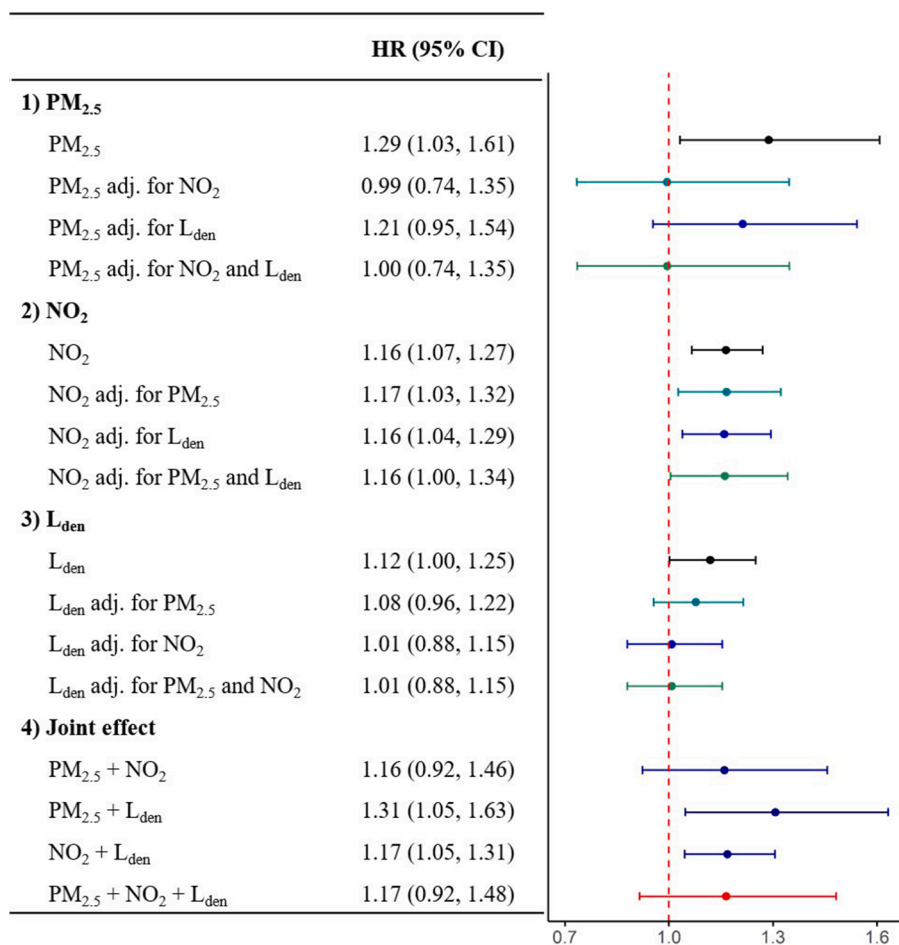


Fig. 3. Associations of single, mutually adjusted, and joint exposure models between 3-year moving average of PM_{2.5}, NO₂, and L_{den} and asthma incidence in the Danish Nurse Cohort. Analyses were based on the fully adjusted models. Estimates are presented as hazard ratios (HRs) and 95% confidence intervals (CIs) based on IQR increases for PM_{2.5} (6.33 µg/m³) and NO₂, (8.19 µg/m³), and per 10 dB increase for L_{den}.

show the most consistent association in mutually adjusted models, which may complicate the interpretation of results in multipollutant models (Butland et al. 2019).

Given that air pollution and road traffic noise exposures occur as a mixture, we attempted to explore the potential for a synergy effect of mixed exposures to air pollutants and road traffic noise, which occurs when the joint effect of combined exposures is greater than the sum of individual exposure effects of the mixture. In our study, we found that the joint HRs based on mutually adjusted models for PM_{2.5}, NO₂, and L_{den} were slightly lower than the combined HRs from single-pollutant models with no interaction effects observed (Table S5), suggesting an antagonistic and confounding effect for the mixed exposures. Further, in the joint effect models with NO₂, the joint HRs were almost equal to the single-pollutant HR for NO₂, suggesting that NO₂ plays the dominant role for adult-asthma incidence, at least with respect to the three exposures considered (PM_{2.5}, NO₂, and L_{den}), and may be sufficient to characterize the toxicity of the exposure mixture in this study.

Biological mechanisms that could link air pollution and road traffic noise exposure to the development of asthma are partly shared and partly different. Exposure to air pollution can induce oxidative stress and activate inflammatory pathways, which may lead to asthma development (Gowers et al. 2012; Kelly 2003). PM_{2.5} can penetrate deep into the respiratory tract and even the lung alveoli and cause oxidative stress, respiratory inflammation, or other biochemical changes related to asthma. Gaseous air pollutants, such as NO₂ and ozone (O₃) as airway irritants, cause oxidative injury to the airways, leading to inflammation, remodelling, and airflow limitation. Annoyance and reduced sleep

quality induced by noise exposure can lead to chronic psychological stress, which is suspected of promoting the development of asthma possibly through immune modulation (Frei et al. 2014; Yonas et al. 2012). Experimental studies suggest that road traffic noise can increase stress hormone levels, thereby triggering inflammatory and oxidative stress pathways (Hahad et al. 2019), which may also contribute to poor respiratory health.

The strengths of our study include prospective cohort design, long follow-up periods, detailed individual information on major covariates, validated state-of-the-art historical estimates of pollution exposures at the residential addresses with information on changes of addresses, and objective assessments of asthma incidence. We benefited from access to road traffic noise exposure information, and provided first data for mutually adjusted models for exposures of PM_{2.5}, NO₂, and L_{den}. A limitation is that information of asthma medication and partial hospital contact before 1977 were not available. We most likely have low sensitivity but high specificity for asthma incidence by using the objective definition (hospital contacts) from the nationwide patient register in Denmark. A Danish study compared the prevalence of asthma using three different classification methods among 7 years old children in the Danish National Birth Cohort, and found that highest prevalence was using prescription data (32.2%) followed by parental self-reports of doctor-diagnosed asthma (12.0%) and registrations of ICD-10 codes in hospital (6.6%) (Stanojevic et al. 2012). However, children, especially the very young ones, are often given medication for asthma-like symptoms, which is different from adults. Defining asthma based on self-reported surveys is subject to recall and information bias, resulting in

less precise ascertainment and potential overestimation of the true asthma burden (Contoli and Papi 2010; Pekkanen and Sunyer 2008). Similarly, using asthma prescription definition can also lead to an overestimation of true asthma burden, as not all of the people using medication had received an asthma diagnosis (Denholm et al. 2020; Zuidgeest et al. 2008). Therefore, the use of hospital contact for asthma definition from a nationwide register is still appealing for epidemiological studies, although it may only indicate a disease progression from a moderate to a more severe stage or asthma exacerbation. A combination of prescription and hospitalization registries for asthma definition would get more complete data of asthma patients, but we did not have access to the prescription registry. Another limitation is that we did not adjust for any area-level confounders for example municipality-level socio-economic status (SES) due to the lack of complete information. However, in our study, the study subjects are all Danish nurses who have similar education and get similar salary regardless of municipality. Therefore, we hypothesize that there is no huge influence of area-level SES on the study results within the same occupation group of nurses. In addition, our exposure assessment relied solely on residential addresses, and we lacked data on work addresses, commuting habits, personal time-activity patterns, and early-life exposure.

5. Conclusion

Our findings from this nationwide nurse cohort in Denmark strengthen the evidence that long-term exposures to ambient air pollution may increase the risk of asthma in adults, and suggest that traffic-related pollutant NO₂ is most relevant. Road traffic noise was not associated with adult-asthma incidence after adjustment for air pollution.

Declaration of Competing Interest

We declare that we have no conflicts of interest.

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Author Contributions

Zorana J. Andersen and Shuo Liu: study conceptualization and design. **Shuo Liu:** statistical analysis and manuscript drafting. **Zorana J. Andersen:** manuscript - review and editing. **Youn-Hee Lim:** methodology and statistical analysis. **Jeanette T. Jørgensen:** cohort data preparation for the analyses. **Heresh Amini:** methodology, data curation, visualization, and writing - review & editing. **Matthias Ketzel, Ole Hertel, Jørgen Brandt, Steen S. Jensen, Jesper H. Christensen, Camilla Geels, and Lise M. Frohn:** developing the exposure modelling system and performing the air pollution exposure calculations. **All authors** have read and revised the manuscript for the important intellectual content, and contributed with the interpretation of the results. **All authors** have approved the final draft of the manuscript.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envint.2021.106464>.

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